

Class Design Document

Team Members:

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Selected Project: Project 8 — Gym Membership & Fitness Class Management System

Class Diagram

A Registration links exactly one Member to exactly one FitnessClass (a many-to-many relationship between Member and FitnessClass, resolved through the Registration "join" class).

Class 1: Member

Owner: Member A

Purpose: Represents one gym member and their personal/membership details.

Attributes

Attribute	Data Type	Description
member_id	str	Unique ID, auto-generated (e.g. "M001")
name	str	Member's full name
age	int	Member's age
membership_type	str	"Basic" or "Premium"
contact	str	Phone number or email, used to prevent duplicates

Methods

Method	Parameters	Purpose
__init__	member_id, name, age, membership_type, contact	Create a new Member
display_details	none	Print the member's info in a readable line
to_row	none	Convert the object to a list of strings for CSV saving
from_row (static)	row	Rebuild a Member object from a CSV row

Class 2: FitnessClass

Owner: Member B

Purpose: Represents one fitness class offered by the gym and tracks capacity.

Attributes

Attribute	Data Type	Description
class_id	str	Unique ID, auto-generated (e.g. "C001")
class_name	str	e.g. "Yoga", "Spin"

Attribute	Data Type	Description
instructor	str	Instructor's name
schedule	str	Day/time, e.g. "Mon 6:00 PM"
capacity	int	Maximum number of members allowed
registered_count	int	Current number of active registrations

Methods

Method	Parameters	Purpose
__init__	class_id, class_name, instructor, schedule, capacity	Create a new class
is_full	none	Return True if registered_count >= capacity
add_registration	none	Increase registered_count by 1
remove_registration	none	Decrease registered_count by 1
display_details	none	Print class info, including OPEN/FULL status
to_row / from_row	—	CSV conversion, same pattern as Member

Class 3: Registration

Owner: Member C

Purpose: Represents one member's sign-up for one fitness class.

Attributes

Attribute	Data Type	Description
registration_id	str	Unique ID, auto-generated (e.g. "R001")
member_id	str	Foreign key to Member.member_id
class_id	str	Foreign key to FitnessClass.class_id
registration_date	str	Date the registration was made (ISO format)
status	str	"Active" or "Cancelled"

Methods

Method	Parameters	Purpose
__init__	registration_id, member_id, class_id, registration_date, status	Create a registration
cancel	none	Set status to "Cancelled"
display_details	none	Print registration info
to_row / from_row	—	CSV conversion, same pattern as Member

Supporting Class: GymSystem (manager)

Purpose: Holds the three lists (members, classes, registrations) and all add/display/search/register/cancel logic and coordinates with file_handler.py for saving and loading. This keeps main.py focused purely on the menu and input validation.

Supporting Module: file_handler.py

Purpose: Pure functions for saving/loading each list to/from a CSV file in the data/ folder, with FileNotFoundError handled gracefully on first run.